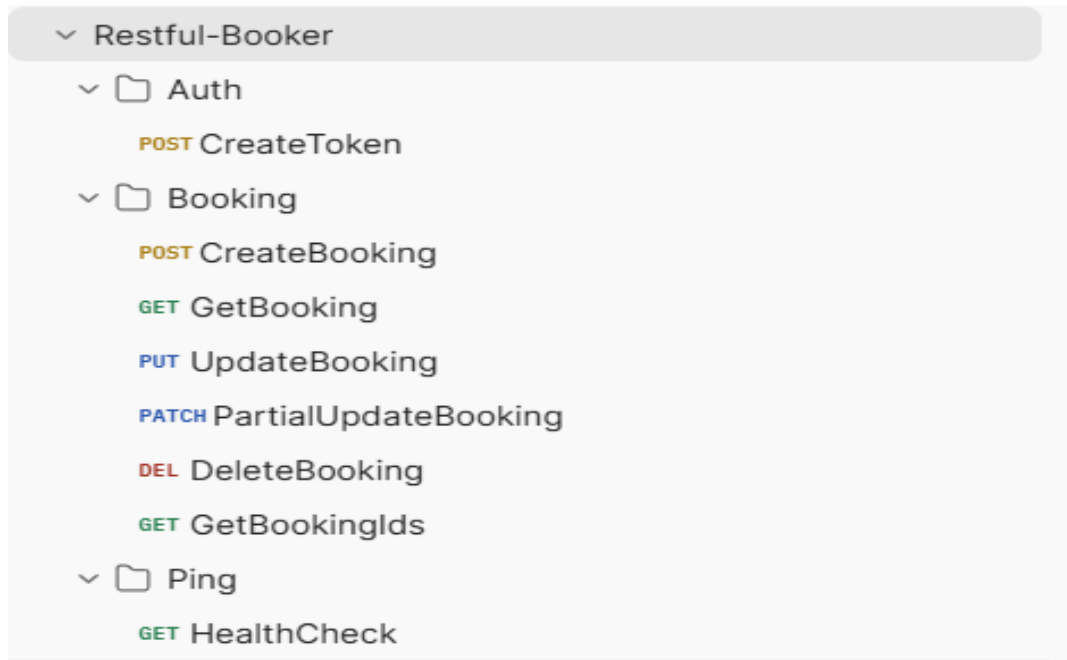


# Restful Booker API

Coverage: Auth | Booking | Health Check

## API Collection Structure :



## Environment Variables

| Variable  | Description            | Example Value                        | Notes                                 |
|-----------|------------------------|--------------------------------------|---------------------------------------|
| base_url  | API base URL           | https://restful-booker.herokuapp.com | Set before running collection         |
| token     | Auth token             | 2207bcab7f68ba7                      | Auto-set after Create Token request   |
| id        | Dynamic booking ID     | 2275                                 | Auto-set after Create Booking request |
| firstname | Test data - first name | Ahmad                                | Used in booking requests              |
| lastname  | Test data - last name  | Test                                 | Used in booking requests              |

# Test Cases Document:

## 1. Introduction

This document contains the complete test suite for the Restful Booker API. It encompasses 34 test cases that cover positive validations, negative validations, error handling, and performance checks across all API endpoints.

## 2. Test Cases

### 1 AUTH – Create Token

| TC ID | Test Case Title                                  | Preconditions         | Test Steps                                             | Expected Result                                                                 | Actual Result                                                                                                          | Status |
|-------|--------------------------------------------------|-----------------------|--------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|--------|
| TC-01 | Validate token generation with valid credentials | API is up and running | 1. Send POST request with valid username and password. | Status code is 200. Token is returned and is not empty. Length > 10 characters. | Status Code: 200 OK. Token returned and is not empty.                                                                  | pass   |
| TC-02 | Validate token performance                       | API is up and running | 1. Send POST request with valid credentials.           | Response time is less than 500 ms.                                              | Validate that response time < 500 ms   AssertionError: expected 607 to be below 500.<br><b>Note: Logged as BUG-01.</b> | fail   |
| TC-03 | Validate token generation with invalid password  | API is up and running | 1. Send POST request with incorrect password.          | Status code is 200. Token is not returned in the response.                      | Status code is 200. Token is not returned.                                                                             | pass   |
| TC-04 | Validate token generation with invalid username  | API is up and running | 1. Send POST request with incorrect username.          | Status code is 200. Token is not returned.                                      | Status code is 200. Token is not returned.                                                                             | pass   |
| TC-05 | Validate token generation with empty credentials | API is up and running | 1. Send POST request with empty request body.          | Status code is 200. Token is not returned                                       | Status code is 200. Token is not returned                                                                              | pass   |

## 2 BOOKING – Create Booking (POST)

### Test Cases

| TC ID | Test Case Title                        | Preconditions         | Test Steps                                                                                     | Expected Result                                         | Actual Result                                                                                                                     | Status |
|-------|----------------------------------------|-----------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------|
| TC-06 | Create booking with valid data         | API is up and running | 1. Send POST request with all required fields (firstname, lastname, totalprice, bookingdates). | Status code is 200. bookingid is returned and not null. | Status Code: 200 OK. bookingid is returned.                                                                                       | pass   |
| TC-07 | Create booking with missing firstname  | API is up and running | 1. Send POST request without the firstname field.                                              | Status code is 400 or 422 or 500.                       | Validate that the request failed and status code is an error code (400, 422, or 500).<br>Validate that no booking ID was returned | pass   |
| TC-08 | Create booking with missing lastname   | API is up and running | 1. Send POST request without the lastname field.                                               | Status code is 400 or 422 or 500.                       | Validate that the request failed and status code is an error code (400, 422, or 500).<br>Validate that no booking ID was returned | pass   |
| TC-09 | Create booking with missing totalprice | API is up and running | 1. Send POST request without the totalprice field.                                             | Status code is 400 or 422 or 500.                       | Validate that the request failed and status code is an error code (400, 422, or 500).<br>Validate that no booking ID was returned | pass   |

| TC ID | Test Case Title                                      | Preconditions         | Test Steps                                                       | Expected Result                    | Actual Result                                                                                                                      | Status |
|-------|------------------------------------------------------|-----------------------|------------------------------------------------------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------|
| TC-10 | Create booking with missing bookingdates             | API is up and running | 1. Send POST request without the bookingdates field.             | Status code is 400 or 422.         | Validate that the request failed and status code is an error code (400, 422, or 500).<br>Validate that no booking ID was returned  | pass   |
| TC-11 | Create booking with invalid date range               | API is up and running | 1. Send POST request where checkout date is before checkin date. | Status code is 400 or 422 or 500.  | Validate that status code is 200.<br><b>Note:<br/>Logged as<br/>BUG-02</b>                                                         | fail   |
| TC-12 | Create booking with invalid data type for totalprice | API is up and running | 1. Send POST request where totalprice is provided as a string.   | Status code is 400 or 422.         | Validate that the request failed and status code is an error code (400, 422, or 500).<br>Validate that no booking ID was returned. | pass   |
| TC-13 | Create booking with empty body                       | API is up and running | 1. Send POST request with an empty body object.                  | Status code is 400 or 500.         | Validate that the request failed and status code is an error code (400, 422, or 500).<br>Validate that no booking ID was returned. | pass   |
| TC-14 | Validate Create Booking Performance                  | API is up and running | 1. Send POST request with valid booking details.                 | Response time is less than 700 ms. | Response time < 700 ms.                                                                                                            | pass   |

### 3 BOOKING – Get Booking (GET) & GetBookingIds (GET).

#### Test Cases:

| TC ID | Test Case Title                           | Preconditions                           | Test Steps                                             | Expected Result                                                 | Actual Result                                             | Status |
|-------|-------------------------------------------|-----------------------------------------|--------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------|--------|
| TC-15 | Retrieve booking by valid ID              | A booking has been successfully created | 1. Send GET request to /booking/{id} using a valid ID. | Status code is 200. Data matches the created booking.           | Status Code: 200 OK. Data matches.                        | pass   |
| TC-16 | Retrieve booking by non-existent ID       | API is up and running                   | 1. Send GET request to /booking/999999.                | Validate that status code is 404 for non-existing booking       | Validate that status code is 404 for non-existing booking | pass   |
| TC-17 | Validate booking structure in response    | A booking has been successfully created | 1. Send GET request to /booking/{id}.                  | Response includes firstname, lastname, and bookingdates object. | Response has structure.                                   | pass   |
| TC-18 | Validate Get Booking performance          | A booking has been successfully created | 1. Send GET request to /booking/{id}.                  | Response time is less than 900 ms.                              | Response time < 900 ms.                                   | pass   |
| TC-19 | Get all booking IDs                       | API is up and running                   | 1. Send GET request to /booking.                       | Status code is 200. Response is returned as an array.           | Status Code: 200 OK. Array returned.                      | pass   |
| TC-20 | Validate structure of get all booking IDs | API is up and running                   | 1. Send GET request to /booking.                       | All objects in the array contain the property bookingid.        | Contains bookingid property.                              | pass   |
| TC-21 | Get bookings with query parameters        | API is up and running                   | 1. Send GET request with a firstname query parameter.  | Status code is 200. Returns relevant booking IDs.               | Status Code: 200 OK. Returns IDs.                         | pass   |

## 4 BOOKING – Update Booking (PUT)

### Test Cases

| TC ID | Test Case Title                          | Preconditions                          | Test Steps                                                  | Expected Result                                             | Actual Result                                                                                   | Status |
|-------|------------------------------------------|----------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------|
| TC-22 | Update booking with valid token and data | Valid token and valid booking ID exist | 1. Send PUT request with valid token and updated body data. | Status code is 200. Old data is replaced with updated data. | Validate that status code is 200 - Update Successful                                            | pass   |
| TC-23 | Update booking without token             | Valid booking ID exists                | 1. Send PUT request without authentication token.           | Status code is 403 or 405.                                  | Validate that status code is 403 or 405 - Forbidden / Method Not Allowed when not authenticated | pass   |
| TC-24 | Update booking with invalid token        | Valid booking ID exists                | 1. Send PUT request with an invalid/expired token.          | Status code is 403 or 405.                                  | Validate that status code is 403 or 405 - Forbidden / Method Not Allowed when not authenticated | pass   |
| TC-25 | Update booking performance               | Valid token and valid booking ID exist | 1. Send PUT request with valid token.                       | Response time is less than 600 ms.                          | Response time < 600 ms.                                                                         | pass   |

## 5 BOOKING – Partial Update (PATCH)

### Test Cases

| TC ID | Test Case Title                         | Preconditions                          | Test Steps                                                  | Expected Result                                                              | Actual Result                                                                                   | Status |
|-------|-----------------------------------------|----------------------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------|
| TC-26 | Partial update booking - firstname only | Valid token and valid booking ID exist | 1. Send PATCH request with valid token to update firstname. | Status code is 200. The firstname is updated without affecting other fields. | Status Code: 200 OK. Validate that firstname is successfully updated.                           | pass   |
| TC-27 | Partial update booking - lastname only  | Valid token and valid booking ID exist | 1. Send PATCH request with valid token to update lastname.  | Status code is 200. The lastname is updated without affecting other fields.  | Status Code: 200 OK. Updated.                                                                   | pass   |
| TC-28 | Partial update booking without token    | Valid booking ID exists                | 1. Send PATCH request without authentication token.         | Status code is 403 or 405.                                                   | Validate that status code is 403 or 405 - Forbidden / Method Not Allowed when not authenticated | pass   |
| TC-29 | Partial update booking performance      | Valid token and valid booking ID exist | 1. Send PATCH request with valid token.                     | Response time is less than 600 ms.                                           | Response time < 600 ms.                                                                         | pass   |

## 6 BOOKING – Delete Booking (DELETE)

### Test Cases

| TC ID | Test Case Title                 | Preconditions                    | Test Steps                                              | Expected Result            | Actual Result                                              | Status |
|-------|---------------------------------|----------------------------------|---------------------------------------------------------|----------------------------|------------------------------------------------------------|--------|
| TC-30 | Delete booking with valid token | Valid token and booking ID exist | 1. Send DELETE request with valid token and booking ID. | Status code is 201.        | Validate that status code is 201 on successful deletion.   | pass   |
| TC-31 | Delete booking without token    | Valid booking ID exists          | 1. Send DELETE request without authentication token.    | Status code is 403 or 405. | Validate that status code is 403 or 405 when missing token | pass   |
| TC-32 | Delete already deleted booking  | Booking was deleted              | 1. Send DELETE request to an already deleted ID.        | Status code is 404 or 405. | 405Method Not Allowed.                                     | pass   |

## 7 HEALTH CHECK – Ping (GET)

### Test Cases

| TC ID | Test Case Title                   | Preconditions         | Test Steps                    | Expected Result                    | Actual Result                                                                                                                        | Status |
|-------|-----------------------------------|-----------------------|-------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------|
| TC-33 | Health Check API availability     | API is up and running | 1. Send GET request to /ping. | Status code is 201.                | Status Code: 201 Created                                                                                                             | pass   |
| TC-34 | Validate health check performance | API is up and running | 1. Send GET request to /ping. | Response time is less than 300 ms. | Validate that response time is less than 300 ms  <br>AssertionError: expected 606 to be below 300.<br><b>Note: Logged as BUG-01.</b> | fail   |



## Bug Report

| Bug ID | Summary                                        | Steps to Reproduce                                                                         | Expected Result                    | Actual Result                         | Severity | Priority |
|--------|------------------------------------------------|--------------------------------------------------------------------------------------------|------------------------------------|---------------------------------------|----------|----------|
| BUG-01 | Token generation response time exceeds 500 ms. | 1. Send POST request to /auth.<br>2. Include valid credentials.<br>3. Check response time. | Response time is less than 500 ms  | Response time is greater than 500 ms. | Minor    | Low      |
| BUG-02 | Invalid date range accepted without error.     | 1. Send POST /booking.<br>2. Set checkout date before checkin date.<br>3. Check response   | 400 Bad Request / validation error | 200 OK - Invalid booking created      | Major    | Medium   |

## • Runner screenshots:

o 1 iteration → basic validation

**Restful-Booker - Run results**

+ New Run Share

Ran today at 12:03:46 AM · [View all runs](#)

| Source | Environment | Iterations | Duration | All tests | Errors | Avg. Resp. Time |
|--------|-------------|------------|----------|-----------|--------|-----------------|
| Runner | booking     | 1          | 3s 375ms | 35        | 0      | 158 ms          |

All 35 Passed 35 Failed 0 Skipped 0 Errors 0 Console log

ListGrid

Iteration 1

**POST** Auth > **CreateToken**  
https://restful-booker.herokuapp.com/auth

200 = 140 ms = 774 B 7

PASS Validate that status code is 200

PASS Validate that response contains token property

PASS Validate that token is not empty

PASS Validate that token is not null

PASS Validate that token is a string

PASS Validate that token length is reasonable (>10 chars)

PASS Validate that response time < 500 ms

**POST** Booking > **CreateBooking**  
https://restful-booker.herokuapp.com/booking

200 = 137 ms = 942 B 5

PASS Validate that status code is 200

PASS Validate that booking id is returned and not null

PASS Validate that the response contains the booking object

PASS Validate that totalprice is a number

PASS Validate that response time is less than 700 ms

**Restful-Booker - Run results**

+ New Run Share

Ran today at 12:03:46 AM · [View all runs](#)

| Source | Environment | Iterations | Duration | All tests | Errors | Avg. Resp. Time |
|--------|-------------|------------|----------|-----------|--------|-----------------|
| Runner | booking     | 1          | 3s 375ms | 35        | 0      | 158 ms          |

All 35 Passed 35 Failed 0 Skipped 0 Errors 0 Console log

ListGrid

https://restful-booker.herokuapp.com/booking/1309

200 = 142 ms = 913 B 5

PASS Validate that status code is 200

PASS Validate that response time is less than 900ms

PASS Validate that response has correct structure and data types

PASS Validate that bookingdates object contains checkin and checkout

PASS Validate that data matches created booking (Environment data)

**PUT** Booking > **UpdateBooking**  
https://restful-booker.herokuapp.com/booking/1309

200 = 140 ms = 919 B 4

PASS Validate that status code is 200 - Update Successful

PASS Validate that response reflects updated values

PASS Validate that other fields remain unchanged

PASS Validate that response time is less than 600 ms

**PATCH** Booking > **PartialUpdateBooking**  
https://restful-booker.herokuapp.com/booking/1309

200 = 155 ms = 919 B 4

PASS Validate that status code is 200

PASS Validate that response time is less than 600 ms

PASS Validate that firstname is successfully updated

PASS Validate that other fields remain unchanged and structure is intact

### DELETE Booking > DeleteBooking

https://restful-booker.herokuapp.com/booking/1309

201 • 139 ms • 747 B 3

- PASS Validate that status code is 201 on successful deletion
- PASS Validate that GET booking returns 404 after deletion
- PASS Validate that deleting already deleted booking returns 404 or 405

### GET Booking > GetBookings

https://restful-booker.herokuapp.com/booking

200 • 273 ms • 16.204 KB 4

- PASS Validate that the status code is 200
- PASS Validate that the response is an array
- PASS Validate that all objects in the array have a bookingid
- PASS Validate that response time is less than 600 ms

### GET Ping > HealthCheck

https://restful-booker.herokuapp.com/ping

201 • 136 ms • 743 B 3

- PASS Validate that status code is 201 - Check API availability
- PASS Validate that response time is less than 300 ms
- PASS Validate that API is responsive and returns 201

## o 8 iterations → advanced testing

### Restful-Booker - Run results

🔄 📺 + 📄 ⌵ Share 🔗 •

🔥 Ran today at 12:06:15 AM • [View all runs](#)

| Source  | Environment | Iterations | Duration  | All tests | Errors      | Avg. Resp. Time |
|---------|-------------|------------|-----------|-----------|-------------|-----------------|
| Runner  | booking     | 8          | 16s 705ms | 280       | 0           | 151 ms          |
| All 280 | Passed 279  | Failed 1   | Skipped 0 | Errors 0  | Console log | List Grid       |

Iteration 1

### POST Auth > CreateToken

https://restful-booker.herokuapp.com/auth

200 • 584 ms • 774 B 6 1

- PASS Validate that status code is 200
- PASS Validate that response contains token property
- PASS Validate that token is not empty
- PASS Validate that token is not null
- PASS Validate that token is a string
- PASS Validate that token length is reasonable (>10 chars)
- FAIL Validate that response time < 500 ms | AssertionError: expected 584 to be below 500

### POST Booking > CreateBooking

https://restful-booker.herokuapp.com/booking

200 • 132 ms • 938 B 5

- PASS Validate that status code is 200
- PASS Validate that booking id is returned and not null
- PASS Validate that the response contains the booking object
- PASS Validate that totalprice is a number
- PASS Validate that response time is less than 700 ms

## Restful-Booker - Run results

     [Share](#)  ...

 Ran today at 12:06:15 AM · [View all runs](#)

| Source                                                     | Environment | Iterations | Duration  | All tests | Errors | Avg. Resp. Time |
|------------------------------------------------------------|-------------|------------|-----------|-----------|--------|-----------------|
| Runner                                                     | booking     | 8          | 16s 705ms | 280       | 0      | 151 ms          |
| All 280 Passed 279 Failed 1 Skipped 0 Errors 0 Console log |             |            |           |           |        |                 |
| List Grid                                                  |             |            |           |           |        |                 |

### GET Booking > GetBooking

<https://restful-booker.herokuapp.com/booking/2683>

200 = 131 ms = 909 B 5

- PASS Validate that status code is 200
- PASS Validate that response time is less than 900ms
- PASS Validate that response has correct structure and data types
- PASS Validate that bookingdates object contains checkin and checkout
- PASS Validate that data matches created booking (Environment data)

### PUT Booking > UpdateBooking

<https://restful-booker.herokuapp.com/booking/2683>

200 = 131 ms = 915 B 4

- PASS Validate that status code is 200 - Update Successful
- PASS Validate that response reflects updated values
- PASS Validate that other fields remain unchanged
- PASS Validate that response time is less than 600 ms

### PATCH Booking > PartialUpdateBooking

<https://restful-booker.herokuapp.com/booking/2683>

200 = 134 ms = 915 B 4

- PASS Validate that status code is 200
- PASS Validate that response time is less than 600 ms
- PASS Validate that firstname is successfully updated
- PASS Validate that other fields remain unchanged and structure is intact

## Restful-Booker - Run results

     [Share](#)  ...

 Ran today at 12:06:15 AM · [View all runs](#)

| Source                                                     | Environment | Iterations | Duration  | All tests | Errors | Avg. Resp. Time |
|------------------------------------------------------------|-------------|------------|-----------|-----------|--------|-----------------|
| Runner                                                     | booking     | 8          | 16s 705ms | 280       | 0      | 151 ms          |
| All 280 Passed 279 Failed 1 Skipped 0 Errors 0 Console log |             |            |           |           |        |                 |
| List Grid                                                  |             |            |           |           |        |                 |

- PASS Validate that other fields remain unchanged and structure is intact

### DELETE Booking > DeleteBooking

<https://restful-booker.herokuapp.com/booking/2683>

201 = 129 ms = 743 B 3

- PASS Validate that status code is 201 on successful deletion
- PASS Validate that GET booking returns 404 after deletion
- PASS Validate that deleting already deleted booking returns 404 or 405

### GET Booking > GetBookingIds

<https://restful-booker.herokuapp.com/booking>

200 = 266 ms = 33.651 KB 4

- PASS Validate that the status code is 200
- PASS Validate that the response is an array
- PASS Validate that all objects in the array have a bookingid
- PASS Validate that response time is less than 600 ms

### GET Ping > HealthCheck

<https://restful-booker.herokuapp.com/ping>

201 = 131 ms = 743 B 3

- PASS Validate that status code is 201 - Check API availability
- PASS Validate that response time is less than 300 ms
- PASS Validate that API is responsive and returns 201

## **Test Summary:**

Total Test Cases: 34 | Sections Covered: 8 | Bug Reports: 2

Runner Sequence: Create Token → Create Booking → Get Booking → Update Booking → Partial Update → Delete Booking → Get Booking (Validation) → Ping.